Ground roll $= S_G = \dfrac{V_{TD}^2}{2g\left[\mu_b - \frac{D}{W}\right]}$ 11

' Limit load

$L_{limit} = n_{limit} \cdot W$

- n_{limit} = limit load factor (g's)

' Ultimate load

$L_{ultimate} = 1.5\, L_{limit}$

$n_{ultimate} = 1.5\, n_{limit}$

' Wing Bending moment

$M_{root} = \dfrac{nWb}{8}$

- W = ~~total lift~~ weight - b = wingspan
= n = load
- $L = nW$

' Gust loads, vertical gust changes load factor

$\Delta n = \dfrac{\rho \cdot V \cdot a \cdot U_{de}}{2(W/S)}$

- a = lift curve slope
- U_{de} = gust velocity

' total load factor becomes $n = 1 + \Delta n$

' Landing gear vertical loads 12

$F = nW$

- Main Gear Load, if main wheel carry 90% of weight then

$F_{main} = 0.9nW$, then each wheel $F_{wheel} = \dfrac{0.9nW}{2}$

- Nose Gear Load

$F_{nose} = 0.1nW$

* End of conceptually designing Aircraft ends HERE *